1. a) Install Flutter and Dart SDK.

Step 1: Install VS Code

Begin by installing VS Code on your computer, if you haven't already.

Step 2: Install Flutter SDK

Go to the official Flutter installation page and download the stable version of the Flutter SDK for your operating system. After downloading, extract the file into a preferred folder, for example, C:\src\flutter.

Step 3: Add Flutter to PATH

Next, you need to add Flutter to your system's PATH environment variables. Copy the path to the bin folder within your Flutter directory (e.g., C:\src\flutter\bin) and add it to your system's Path. After this, make sure to restart your computer.

Step 4: Install Extensions in VS Code

Open VS Code, go to the Extensions tab (Ctrl+Shift+X), and search for and install the "Flutter" and "Dart" extensions.

Step 5: Check Installation

verify that everything is set up correctly, open the terminal in VS Code and type flutter doctor. If the installation is successful, you will see a checkmark (✔) next to the installed components. If anything is missing, the command will provide instructions on how to fix it.

Step 6: Create & Run Flutter Project

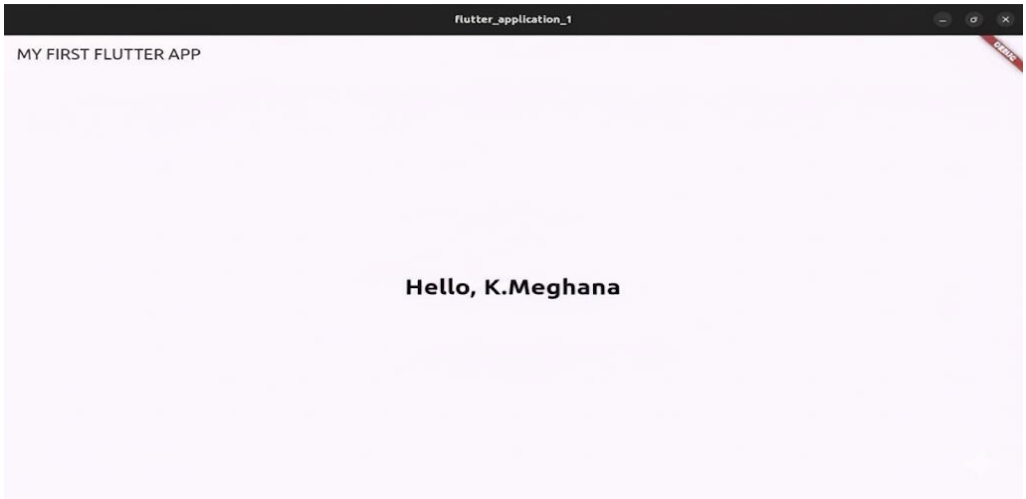
Finally, you can create and run your first Flutter project. In the VS Code terminal, type flutter create my_app, then navigate into the new directory with cd my_app. You can then open the project in VS Code with code . and run it by pressing F5 or clicking "Run" in VS Code.

1.b) Write a simple Dart program to understand the language basics.

```
import 'package:flutter/material.dart';

void main() {

  runApp(MyApp());

}

class MyApp extends StatelessWidget {

  @override
```

```
Widget build(BuildContext context) {
  String name = "K.Meghana"; // Your name
  return MaterialApp(
    home: Scaffold(
      appBar: AppBar(
        title: Text('MY FIRST FLUTTER APP'),
      ),
      body: Center(
        child: Text(
          "Hello, $name",
          style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold),
        ),
  );
}
```

OUTPUT :

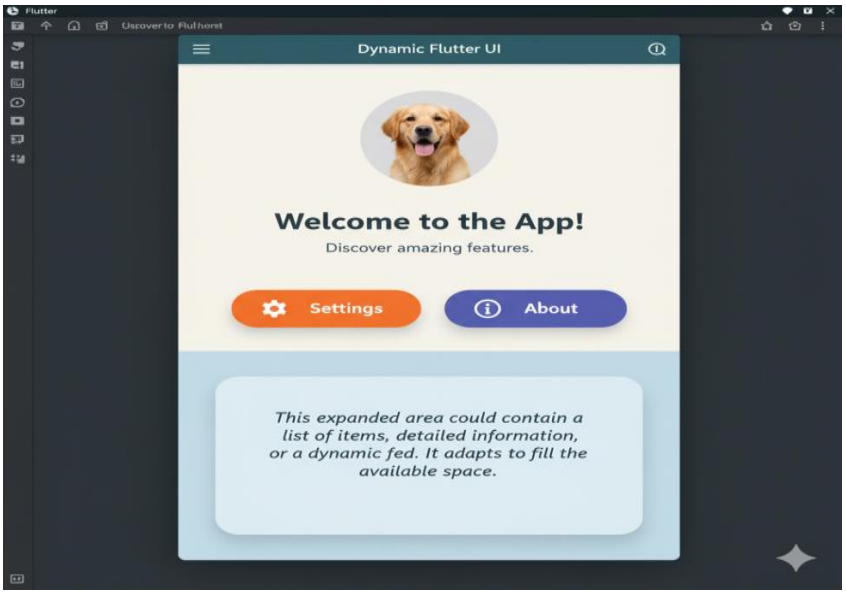


2.a) Explore various Flutter widgets (Text, Image, Container, etc.).

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Dynamic Flutter UI',
      home: Scaffold(
        appBar: AppBar(
          title: Text('Dynamic Flutter UI'),
          backgroundColor: Colors.teal[700],
          elevation: 0,
        ),
        body: Column(
          children: [
              children: [
                CircleAvatar(
                  radius: 50,
                  backgroundImage: NetworkImage('https://picsum.photos/id/1025/200/200'), // Example image
                  backgroundColor: Colors.grey[200],
                ),
                SizedBox(height: 15),
                Text(
                  'Welcome to the App!',
                  style: TextStyle(
                ),
                SizedBox(height: 10),
                Text(
                  'Discover amazing features.',
                  style: TextStyle(fontSize: 16, color: Colors.blueGrey[600]),
              ],
            ),
          ),
          Padding(
            padding: const EdgeInsets.symmetric(horizontal: 20.0, vertical: 15.0),
            child: Row(
              mainAxisAlignment: MainAxisAlignment.spaceAround,
              children: [
```

```
                ElevatedButton.icon(
                  onPressed: () {},
                  icon: Icon(Icons.settings),
                  label: Text('Settings'),
                  style: ElevatedButton.styleFrom(
                      borderRadius: BorderRadius.circular(10),
                ),
                ElevatedButton.icon(
                  onPressed: () {},
                  icon: Icon(Icons.info),
                  label: Text('About'),
                  style: ElevatedButton.styleFrom(
                    shape: RoundedRectangleBorder(
                      borderRadius: BorderRadius.circular(10),
                    ),
              ],
            ),
          ),
          Expanded(
            child: Padding(
              padding: const EdgeInsets.all(20.0),
              child: Card(
                elevation: 8,
                shape: RoundedRectangleBorder(
                  borderRadius: BorderRadius.circular(15),
                ),
        ],
      ),
    ),
  );
}
}
```

OUTPUT:

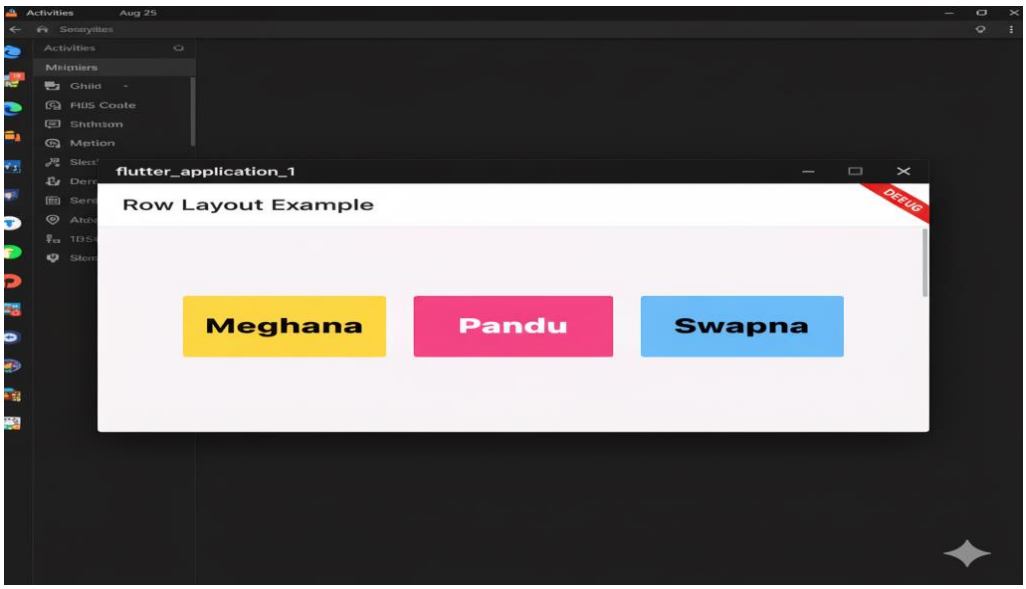


b) Implement different layout structures using Row, Column, and Stack widgets.
Row:

```
import 'package:flutter/material.dart';

void main() {

  runApp(MyApp());

}

class MyApp extends StatelessWidget {

  @override

  Widget build(BuildContext context) {

    return MaterialApp(

      home: Scaffold(

        appBar: AppBar(

          title: Text('Row Layout Example'),

        ),

        body: Center(

          child: Row(

            mainAxisAlignment: MainAxisAlignment.spaceEvenly, // Distribute space evenly

            children: <Widget>[

              Container(

                width: 100,

                height: 100,

                color: Colors.yellow, // Color for Meghana

                child: Center(

                  child: Text(

                    'Meghana',

                    style: TextStyle(color: Colors.black, fontWeight: FontWeight.bold),
```

```
        Container(
          width: 100,
          height: 100,
          color: Colors.pinkAccent, // Color for Pandu
        ),
        Container(
          width: 100,
          height: 100,
          color: Colors.lightBlue, // Color for Swapna
          child: Center(
            child: Text(
              'Swapna',
              style: TextStyle(color: Colors.white, fontWeight: FontWeight.bold),
        ),
      ],
    ),
  );
}
}
```

OUTPUT:



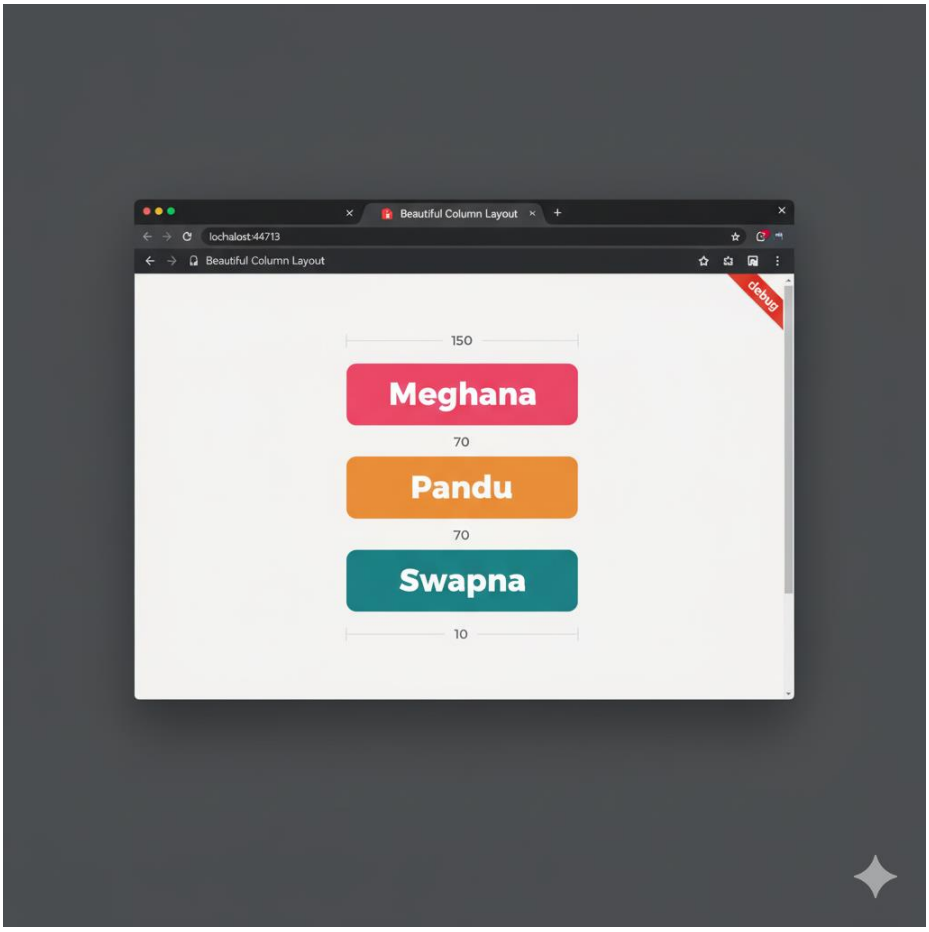
Column :

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: Text('Beautiful Column Layout'),
          backgroundColor: Colors.deepPurple,
        ),
        body: Center(
            Container(
              width: 150,
              height: 70,
              margin: EdgeInsets.symmetric(vertical: 10), // Add vertical spacing
              decoration: BoxDecoration(
                color: Colors.pinkAccent, // Color for Meghana
                borderRadius: BorderRadius.circular(15), // Rounded corners
              ),
              child: Center(
                child: Text(
```

```
      'Meghana',
      style: TextStyle(color: Colors.white, fontWeight: FontWeight.bold, fontSize: 18),
    ),
  ),
),
Container(
  width: 150,
  decoration: BoxDecoration(
    color: Colors.orange, // Color for Pandu
    borderRadius: BorderRadius.circular(15),
  ),
  child: Center(
    child: Text(
      'Pandu',
      style: TextStyle(color: Colors.white, fontWeight: FontWeight.bold, fontSize: 18),
    ),
  ),
),
Container(
  width: 150,
  decoration: BoxDecoration(
    color: Colors.teal, // Color for Swapna
    borderRadius: BorderRadius.circular(15),
  ),
  child: Center(
```

```
                child: Text(
                  'Swapna',
                  style: TextStyle(color: Colors.white, fontWeight: FontWeight.bold, fontSize: 18),
              ),
            ],
    );
  }
}
```

OUTPUT :



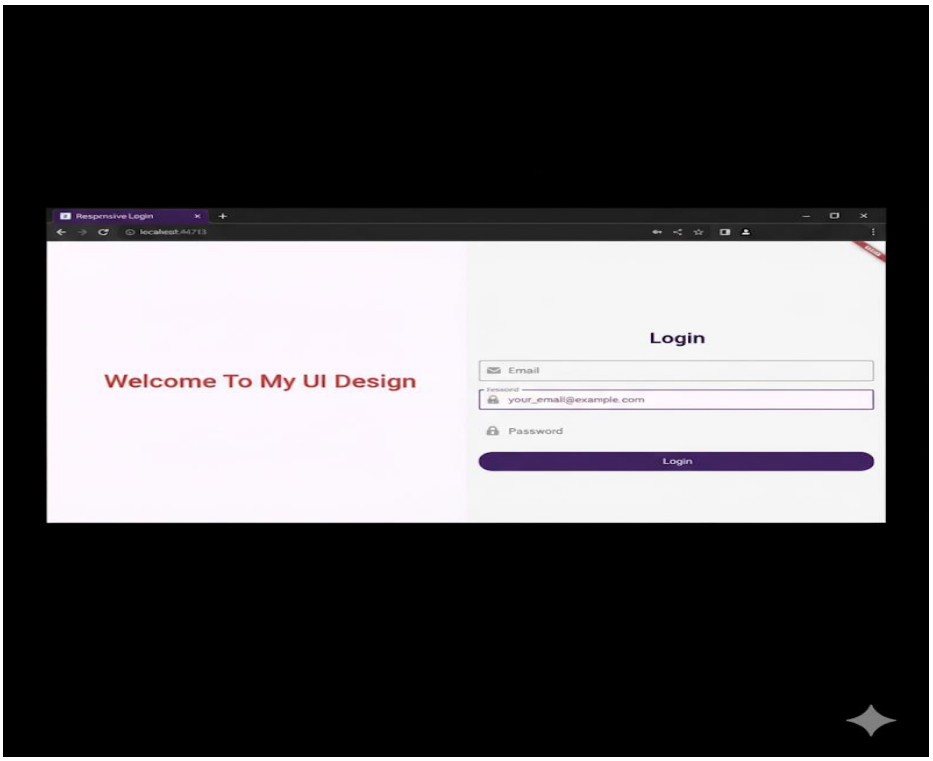
3.a) Design a responsive UI that adapts to different screen sizes.

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Responsive Login',
      theme: ThemeData(
        primarySwatch: Colors.deepPurple,
      ),
      home: ResponsiveLoginPage(),
    );
class ResponsiveLoginPage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Responsive Login'),
      ),
          return Row(
            children: [
              Expanded(
```

```
            Expanded(
              flex: 1,
              child: LoginForm(),
        } else {
          child: Column(
            children: [
              Container(
                child: Text(
                  'Welcome To My UI Design',
                  textAlign: TextAlign.center,
                  style: TextStyle(
                    fontSize: 28, // Smaller font for mobile
                    fontWeight: FontWeight.bold,
                    color: Colors.red[800],
              ),
              LoginForm(),
        ],
  }
}
class LoginForm extends StatefulWidget {
  @override
  _LoginFormState createState() => _LoginFormState();
}
class _LoginFormState extends State<LoginForm> {
  final TextEditingController _emailController = TextEditingController(text: 'your_email@example.com');
```

```
            textStyle: TextStyle(fontSize: 18),
          ),
          child: Text('Login'),
        ),
      ],
    ),
  );
 }
}
```

OUTPUT :



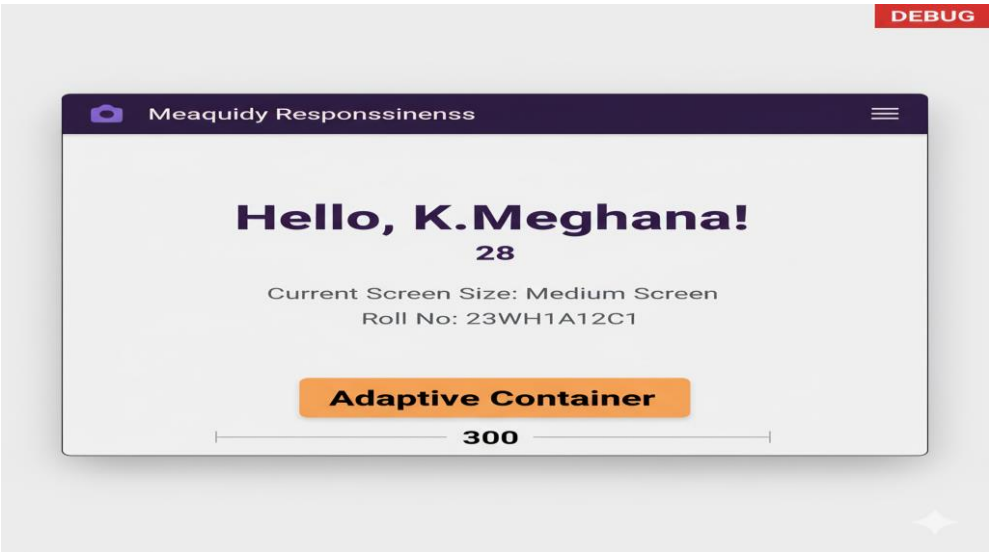
3.b) Implement media queries and breakpoints for responsiveness.

```
import 'package:flutter/material.dart';


void main() {
  runApp(MyApp());
}


class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: ResponsiveTextPage(),
    );
  }
}class ResponsiveTextPage extends StatelessWidget {
  @override
  Widget build(BuildContext context) {    final bool isLargeScreen = screenWidth > 800;
    final bool isMediumScreen = screenWidth > 400 && screenWidth <= 800;
    double fontSize;
    String screenType;    if (isLargeScreen) {
      fontSize = 40.0;
      screenType = "Large Screen";
else {
      fontSize = 20.0;
```

```
  screenType = "Small Screen";
}
  body: Center(
    child: Column(
      mainAxisAlignment: MainAxisAlignment.center,
      children: [
        Text(
          'Hello, K.Meghana!',
          style: TextStyle(
            fontSize: fontSize,
            fontWeight: FontWeight.bold,
            color: Colors.deepPurple,
          ),
          textAlign: TextAlign.center,
        ),
        SizedBox(height: 20),
        Text(
          'Current Screen Size: $screenType',
          style: TextStyle(fontSize: 18, color: Colors.grey[700]),
        ),
        SizedBox(height: 10),
        Text(
          'Roll No: 23WH1A12C1',
          style: TextStyle(fontSize: 16, color: Colors.grey[600]),
```

```
          ),
           SizedBox(height: 30),
          Container(
            width: isLargeScreen ? 600 : (isMediumScreen ? 300 : 200),
            height: 50,
            color: isLargeScreen ? Colors.greenAccent : (isMediumScreen ? Colors.orangeAccent : Colors.redAccent),
            child: Center(child: Text('Adaptive Container', style: TextStyle(color: Colors.black, fontWeight:
FontWeight.bold))),
          )
        ],
      ),
    ),
  );
 }
}
```

OUTPUT:



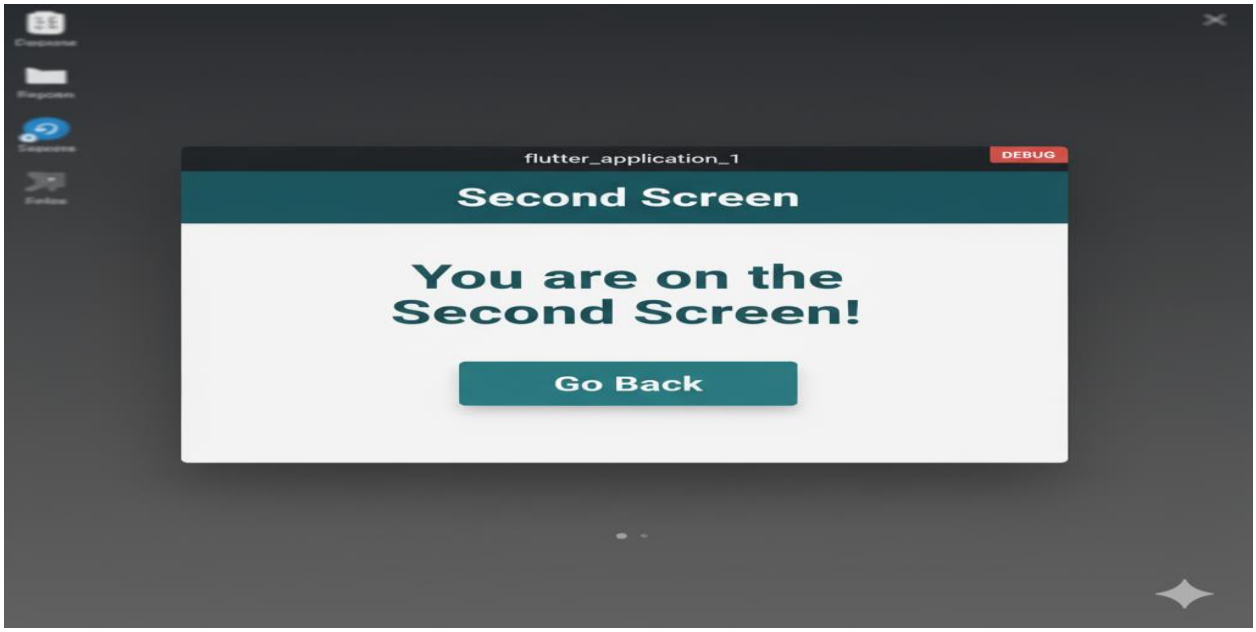
4. a) Set up navigation between different screens using Navigator.

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: FirstScreen(),
class FirstScreen extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('First Screen'),
        backgroundColor: Colors.blueGrey,
      ),
      body: Center(
        child: Column(
              onPressed: () {
                Navigator.push(
                  context,
                  MaterialPageRoute(builder: (context) => SecondScreen()),
                );
```

```
            },
            child: Text('Go to Second Screen'),
            style: ElevatedButton.styleFrom(
              padding: EdgeInsets.symmetric(horizontal: 25, vertical: 15),
            ),
          ),
        ],
  }
}class SecondScreen extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Second Screen'),
        backgroundColor: Colors.teal,
      ),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
            onPressed: () {
              Navigator.pop(context); // Go back to the previous screen
            },
            child: Text('Go Back'),
            style: ElevatedButton.styleFrom(
              backgroundColor: Colors.teal[700],
```

```
            foregroundColor: Colors.white,
            padding: EdgeInsets.symmetric(horizontal: 25, vertical: 15),
        ],
    );
  );
 }
}
```

OUTPUT :



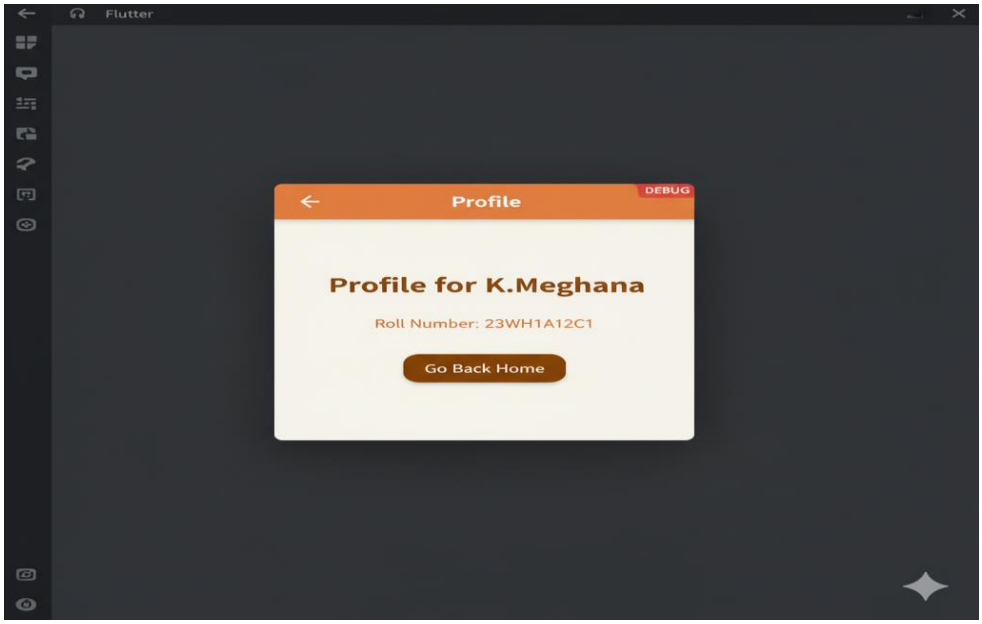
4. b) Implement navigation with named routes.

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      initialRoute: '/',
      routes: {
        '/': (context) => HomeScreen(),
        '/profile': (context) => ProfileScreen(),
      },
    );
  }
}
class HomeScreen extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Home (Named Routes)'),
        backgroundColor: Colors.purple,
      ),
```

```
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Text('Hello, K.Meghana!', style: TextStyle(fontSize: 22, color: Colors.purple[800])),
            SizedBox(height: 20),
                padding: EdgeInsets.symmetric(horizontal: 30, vertical: 15),
              ),
            ),
          ],
  }
}
class ProfileScreen extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            Text(
              'Profile for K.Meghana',
              style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold, color: Colors.orange[800]),
            ),
            SizedBox(height: 10),
            Text(
              'Roll Number: 23WH1A12C1',
```

```
            style: TextStyle(fontSize: 18, color: Colors.orange[600]),
          ),
          SizedBox(height: 30),
          ElevatedButton(
            onPressed: () {
              Navigator.pop(context); // Go back to the home screen
            },
              padding: EdgeInsets.symmetric(horizontal: 25, vertical: 15),
            ),
          ),
        ],
    );
  }
}
```

OUTPUT :



5. a) Design stateful and stateless widgets.

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: Text('Stateful vs Stateless'),
          backgroundColor: Colors.blueGrey[800],
        ),
        body: Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: <Widget>[
          }
class MyInfoWidget extends StatelessWidget {
  final String name;
  final String rollNumber;
  MyInfoWidget({required this.name, required this.rollNumber});
  @override
  Widget build(BuildContext context) {
```

```
    return Container(
      padding: EdgeInsets.all(20),
        borderRadius: BorderRadius.circular(10),
        boxShadow: [BoxShadow(color: Colors.grey.withOpacity(0.3), blurRadius: 5)],
      ),
      child: Column(
        children: [
          Text(
            'Static Info (Stateless)',
            style: TextStyle(fontSize: 20, fontWeight: FontWeight.bold, color: Colors.blueGrey[700]),
          ),
          SizedBox(height: 10),
          Text('Name: $name', style: TextStyle(fontSize: 16)),
          Text('Roll No: $rollNumber', style: TextStyle(fontSize: 16)),
        ]
class MyCounterWidget extends StatefulWidget {
  @override
  _MyCounterWidgetState createState() => _MyCounterWidgetState();
}
class _MyCounterWidgetState extends State<MyCounterWidget> {
  int _counter = 0;


  void _incrementCounter() {
    setState(() {
      _counter++; // Update the state
```

```
  });
 @override
 Widget build(BuildContext context) {
  return Container(
    padding: EdgeInsets.all(20),
      boxShadow: [BoxShadow(color: Colors.grey.withOpacity(0.3), blurRadius: 5)],
    ),
      SizedBox(height: 10),
      ElevatedButton(
        style: ElevatedButton.styleFrom(
          backgroundColor: Colors.green[700],
          foregroundColor: Colors.white,
          padding: EdgeInsets.symmetric(horizontal: 20, vertical: 10),
      ],
  );
 }
}
```

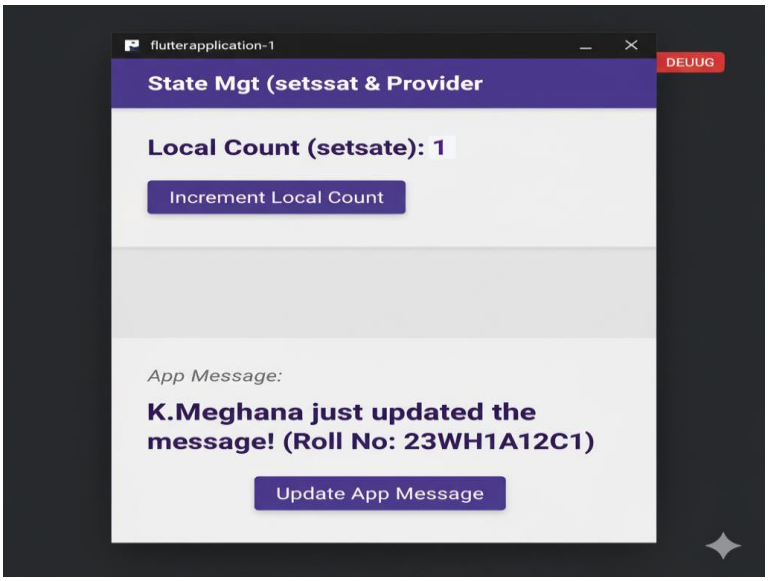
OUTPUT :

5. b) Implement state management using `setState` and `Provider`.

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';
void main() {
  runApp(
    ChangeNotifierProvider(
      create: (context) => AppMessageModel(),
      child: MyApp(),
    ),
  );
class AppMessageModel extends ChangeNotifier {
  String _message = 'Initial message from K.Meghana';
    void updateMessage(String newMessage) {
    _message = newMessage;
    notifyListeners(); // Notify all listeners to rebuild
  }
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: StateManagementExample(),
    );
  }
class StateManagementExample extends StatefulWidget {
```

```
        padding: const EdgeInsets.all(8.0),
        child: Text(
          appMessage,
          textAlign: TextAlign.center,
          style: TextStyle(fontSize: 24, fontWeight: FontWeight.bold, color: Colors.deepPurple[700]),
        ),
      ),
      SizedBox(height: 20),
      ElevatedButton(
        onPressed: () {
            .updateMessage('K.Meghana just updated the message! (Roll No: 23WH1A12C1)');
        },
        child: Text('Update App Message'),
        style: ElevatedButton.styleFrom(backgroundColor: Colors.deepPurpleAccent),
      ),
    );
  }
}
```

OUTPUT :



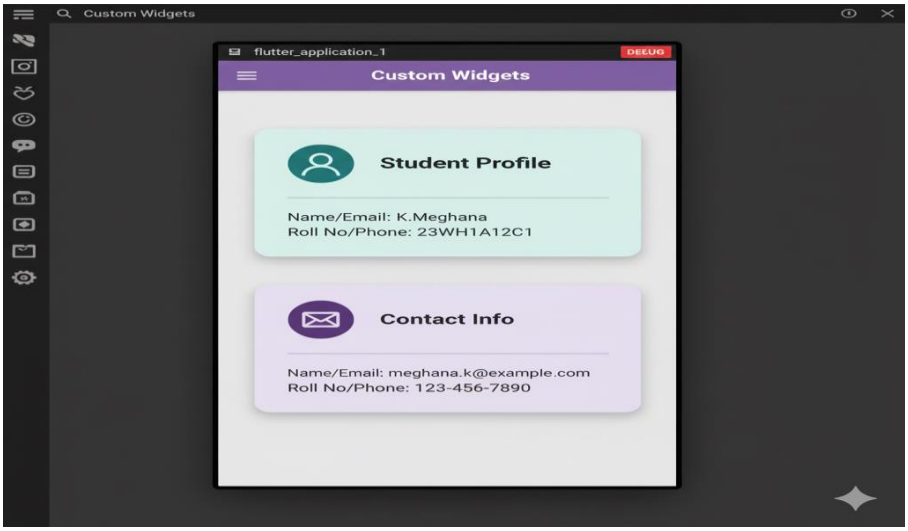
6. a) Create custom widgets for UI elements.

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: Text('Custom Widgets'),
          backgroundColor: Colors.teal,
        ),
        body: Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
              CustomDetailCard(
                title: 'Student Profile',
                name: 'K.Meghana',
                rollNumber: '23WH1A12C1',
                icon: Icons.person,
                cardColor: Colors.teal[100]!,
                iconColor: Colors.teal[700]!,
              ),
```

```
                CustomDetailCard(
                  title: 'Contact Info',
                  name: 'meghana.k@example.com',
                  rollNumber: '123-456-7890', // Just an example number
                  icon: Icons.contact_mail,
                  cardColor: Colors.purple[100]!,
                  iconColor: Colors.purple[700]!,
                ),
              ],
      ),
    );
  }
}
class CustomDetailCard extends StatelessWidget {
  final String title;
  final Color cardColor;
  final Color iconColor;
  @override
  Widget build(BuildContext context) {
    return Card(
      elevation: 5,
      shape: RoundedRectangleBorder(borderRadius: BorderRadius.circular(15)),
      margin: EdgeInsets.symmetric(horizontal: 20),
      color: cardColor,
      child: Padding(
```

```
        padding: const EdgeInsets.all(20.0),
        child: Column(
          ],
        ),
        SizedBox(height: 15),
        Divider(color: Colors.grey[400]),
        SizedBox(height: 10),
        Text(
          'Name/Email: $name',
          style: TextStyle(fontSize: 16, color: Colors.black54),
        ),
        SizedBox(height: 5),
    );
  }
}
```

OUTPUT :

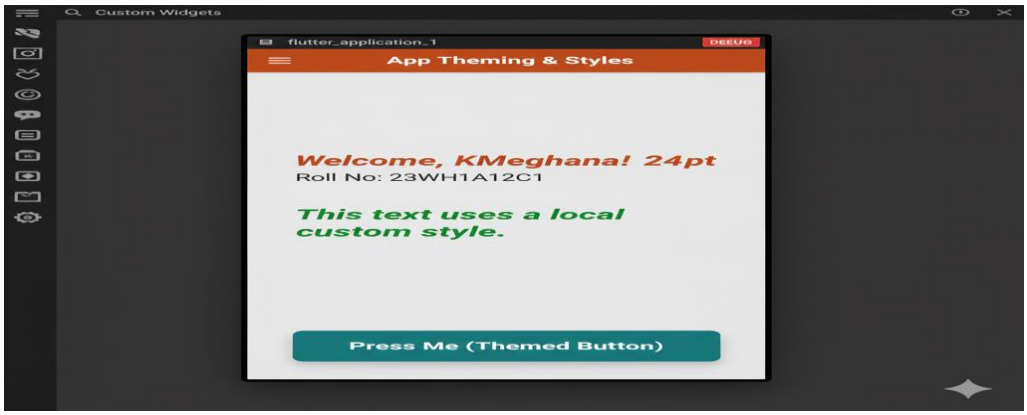


6.b) Apply styling using themes and custom styles.

```
import 'package:flutter/material.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Theming & Styles',
        colorScheme: ColorScheme.fromSwatch(
          primarySwatch: Colors.deepOrange, // Primary color for the app
        ).copyWith(
          secondary: Colors.teal
            fontWeight: FontWeight.bold,
            fontFamily: 'Roboto', // Custom font if loaded
          ),
        ),
        elevatedButtonTheme: ElevatedButtonThemeData(
          style: ElevatedButton.styleFrom(
            backgroundColor: Colors.teal[600], // Teal buttons
            foregroundColor: Colors.white,
            padding: EdgeInsets.symmetric(horizontal: 30, vertical: 15),
            shape: RoundedRectangleBorder(
              borderRadius: BorderRadius.circular(10),
```

```
      style: theme.textTheme.bodyMedium,
    ),
    SizedBox(height: 30),
    ElevatedButton(
      onPressed: () {
        // Button uses the ElevatedButtonThemeData defined in MaterialApp
        ScaffoldMessenger.of(context).showSnackBar(
          SnackBar(
            content: Text('Styled Button Pressed!'),
            backgroundColor: theme.colorScheme.secondary,
          ),
        );
      },
      child: Text('Press Me (Themed Button)'),
    ),
  ],
}
```

OUTPUT :



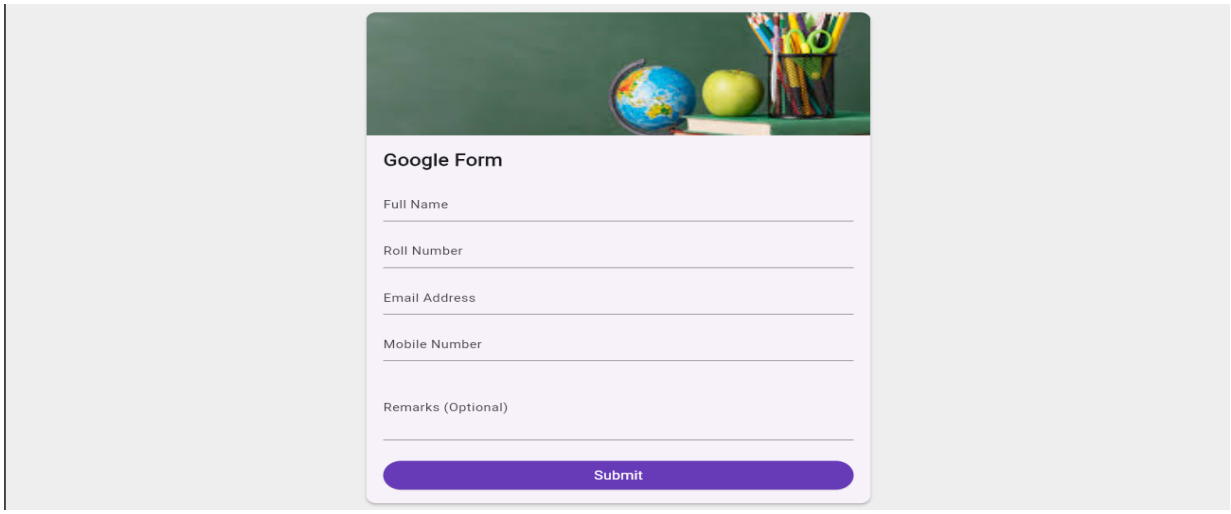
7. a) Design a form with various input fields.

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: const RegistrationForm(),
    );
  }
class _RegistrationFormState extends State<RegistrationForm> {
  final _formKey = GlobalKey<FormState>();


  final nameController = TextEditingController();
  void _submitForm() {
    if (_formKey.currentState!.validate()) {
      ScaffoldMessenger.of(context).showSnackBar(
        const SnackBar(content: Text("Form Submitted Successfully!")),
      );
    }
  }
  @override
  Widget build(BuildContext context) {
```

```
    border: UnderlineInputBorder(),
  ),
const SizedBox(height: 20),
TextFormField(
  controller: emailController,
  decoration: const InputDecoration(
    labelText: "Email Address",
    border: UnderlineInputBorder(),
  ),
  keyboardType: TextInputType.emailAddress,
  validator: (value) {
    if (value == null || value.isEmpty) {
      return "Enter your email";
    }
    if (!RegExp(r'^[^@]+@[^@]+\.[^@]+')
        .hasMatch(value)) {
      return "Enter valid email";
    }
    return null;
  },
),
    if (!RegExp(r'^[0-9]{10}$').hasMatch(value)) {
      return "Enter valid 10-digit number";
    }
    return null;
```

```
                    },
                  ),
                  ElevatedButton(
                    style: ElevatedButton.styleFrom(
                      backgroundColor: Colors.deepPurple,
                      padding:
                        const EdgeInsets.symmetric(vertical: 16),
                    ),
                ),
              ),
            ),
          ],
        ),
}
```

OUTPUT :



7.b) Implement form validation and error handling.

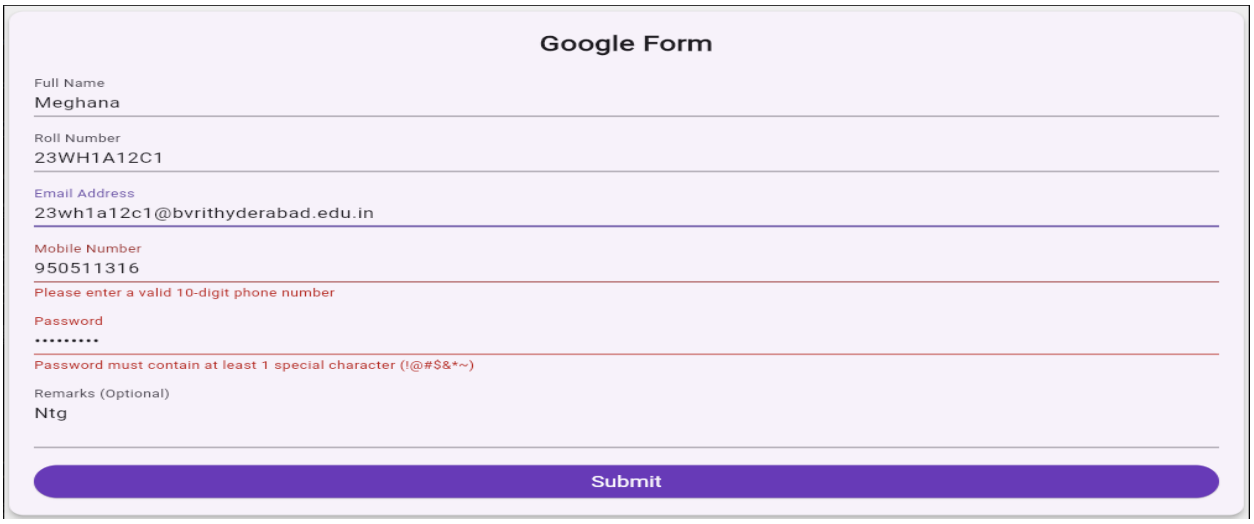
```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: "Form with Validations",
      theme: ThemeData(primarySwatch: Colors.deepPurple),
      home: const RegistrationForm(),
    );
  }
}
class RegistrationForm extends StatefulWidget {
  const RegistrationForm({super.key});
  @override
  State<RegistrationForm> createState() => _RegistrationFormState();
}
  final TextEditingController mobileController = TextEditingController();
  final TextEditingController passwordController = TextEditingController();
  final TextEditingController remarksController = TextEditingController();
```

```
void _submitForm() {
  if (_formKey.currentState!.validate()) {
    ScaffoldMessenger.of(context).showSnackBar(
      const SnackBar(content: Text("□ Form submitted successfully!")),
    );
  } else {
    ScaffoldMessenger.of(context).showSnackBar(
      const SnackBar(content: Text("□ Please fix the errors")),
    );
String? _validatePassword(String? value) {
  if (value == null || value.isEmpty) {
    return "* This field cannot be empty";
  }
  if (value.length < 8) {
    return "Password must be at least 8 characters long";
  }
  if (!RegExp(r'[A-Z]').hasMatch(value)) {
    return "Password must contain at least 1 uppercase letter";
  }
  if (!RegExp(r'[a-z]').hasMatch(value)) {
    return "Password must contain at least 1 lowercase letter";
  }
  if (!RegExp(r'[0-9]').hasMatch(value)) {
    return "Password must contain at least 1 digit";
  }
```

```
  if (!RegExp(r'[!@#\$&*~]').hasMatch(value)) {
    return "Password must contain at least 1 special character (!@#\$&*~)";
  }
  return null;
}
@override
Widget build(BuildContext context) {
  return Scaffold(
    backgroundColor: Colors.grey[200],
    body: Center(
        child: Padding(
          padding: const EdgeInsets.all(20.0),
          child: Form(
                decoration: const InputDecoration(
                  labelText: "Full Name",
                  border: UnderlineInputBorder(),
                ),
               TextFormField(
                controller: rollController,
                decoration: const InputDecoration(
                  labelText: "Roll Number",
                  border: UnderlineInputBorder(),
                ),
                validator: (value) {
                  if (value == null || value.isEmpty) {
```

```
                ElevatedButton(
                  style: ElevatedButton.styleFrom(
                    shape: RoundedRectangleBorder(
                      borderRadius: BorderRadius.circular(25),
                    ),
                  ),
                  onPressed: _submitForm,
                  child: const Text(
                    "Submit",
                    style: TextStyle(fontSize: 18, color: Colors.white),
                  ),
            ),
    );
  }
}
```

OUTPUT :



8. a) Add animations to UI elements using Flutter's animation framework.

```
import 'package:flutter/material.dart';

void main() {

  runApp(const MyApp());

}

class MyApp extends StatelessWidget {

  const MyApp({super.key});

  @override

  Widget build(BuildContext context) {

    return MaterialApp(

      debugShowCheckedModeBanner: false,

      title: 'Image Carousel',

    );

  }

}


class ImageCarousel extends StatefulWidget {

  const ImageCarousel({super.key});

  @override

  State<ImageCarousel> createState() => _ImageCarouselState();

}

  final List<String> images = [

    "assets/tom1.jpeg",

    "assets/tom2.png",

    "assets/tom3.jpeg",

  ];
```

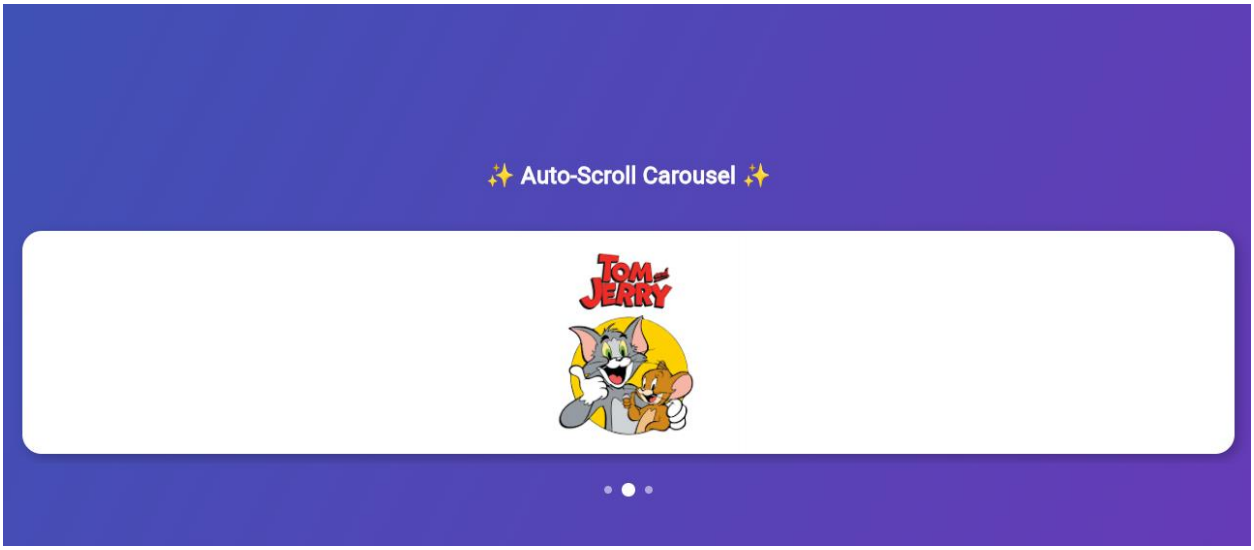
```
  style: TextStyle(
    fontSize: 24,
    fontWeight: FontWeight.bold,
    color: Colors.white,
  ),
),
SizedBox(
  height: 250,
  child: PageView.builder(
    controller: _pageController,
        boxShadow: const [
          BoxShadow(
            color: Colors.black26,
            blurRadius: 10,
            offset: Offset(4, 4),
          ),
        ],
      ),
      child: ClipRRect(
        borderRadius: BorderRadius.circular(20),
        child: Image.asset(
          images[index],
          fit: BoxFit.contain, // □ shows full image neatly
        ),
  ),
```

```
        ),
      ),
    );
  }
}
```

OUTPUT :

8. b) Experiment with different types of animations (fade, slide, etc.).

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp
}
class ImageCarousel extends StatefulWidget {
  const ImageCarousel({super.key});
  @override
  State<ImageCarousel> createState() => _ImageCarouselState();
}
  final List<String> images = [
    "assets/tom1.jpeg",
    "assets/tom2.png",
    "assets/tom3.jpeg",
  ];
  void _startAutoScroll() {
    Future.delayed(const Duration(seconds: 3), () {
      if (_pageController.hasClients) {
        setState(() {
```

```
            color: Colors.white,
          ),
        ),
        Row(
          mainAxisAlignment: MainAxisAlignment.center,
          children: List.generate(images.length, (index) {
            return AnimatedContainer(
              height: _currentPage == index ? 14 : 8,
              decoration: BoxDecoration(
                color:
      ),
    ),
  );
 }
}
```

OUTPUT: